



Cross-Vendor MCU Board Comparison

Dkhili Fares

Cross-vendor MCU Benchmarking

Objective: Define fair, technical comparisons between multiple MCU evaluation board and their closest STM32 counterparts.

Focus:

- Core performance and execution time
 - Flash/RAM footprint
 - Peripherals & ecosystem
 - Middleware / SDKs



Boards in Scope (Overview)

NXP

- MIMXRT1060-EVKC (i.MX RT1062 – M7 crossover)
- FRDM-MCXA156 (M33 low-power/industrial)

Renesas

- FPB-RA0E1 (M23, low cost)
- FPB-RA8E1 (M85, high performance)

TI

- LP-MSPM0C1104 (M0+, ultra-low-end)

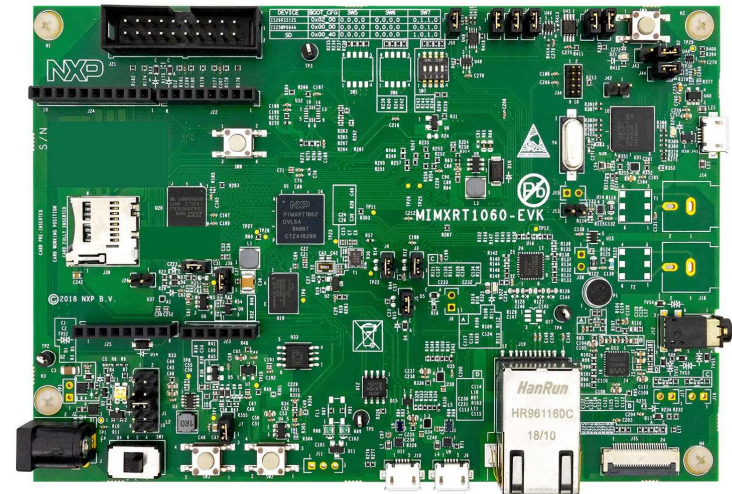
Gega Device

- GD32E5072Z-EVAL



NXP: MIMXRT1060-EVKC (i.MX RT1062 – M7 crossover)

The MIMXRT1060-EVKC board is an evaluation and development platform for the NXP i.MX RT1060 crossover MCU, which uses the Arm Cortex-M7 core. The i.MX RT1060 MCU is optimized for real-time performance and high integration for industrial and IoT applications.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

1

Core and clock system

Criterion	NXP board – MIMXRT1060-EVKC	STM32 board – STM32H735G-DK	Tag	Reason
Core type	Arm Cortex-M7	Arm Cortex-M7	near	Same architecture (M7, same ISA, FPU), ideal for execution-time comparison.
Max core frequency	Up to **600 MHz**	Up to **550 MHz**	near	Both are high-end M7; frequency difference is moderate and can be normalized.
Internal code Flash	**None**: flashless MCU, code in external QSPI/HyperFlash	**2 MB on-chip Flash**	not near	RT1060 is a crossover MCU relying on external Flash; H735 has large on-chip Flash.
Internal SRAM	≈**1 MB** tightly-coupled / on-chip RAM (ITCM/DTCM + OCRAM)	≈**1 MB+** on-chip SRAM (ITCM/DTCM + AXI SRAM)	near	Both provide about 1 MB of fast on-chip RAM for code/data; good for fair RAM benchmarks.
External memory IF	**SEMC** (SDRAM/NAND) + **FlexSPI** (QSPI/HyperFlash)	**FMC** (SDRAM) + **QUADSPI**	near	Different names, same roles: parallel SDRAM controller + serial NOR Flash controller.
External memories	**32 MB SDRAM**, **8 MB QSPI NOR**, **64 MB HyperFlash option**	**External SDRAM** (on DK) + **QSPI NOR Flash**	near	Both offer external SDRAM + serial Flash; RT1060 has HyperFlash but functionally comparable.
Package / pins	MIMXRT1062DVL6B, high-pin-count LQFP/BGA	STM32H735IGK6, LQFP144	not near	H735 has fewer pins; overall I/O budget is smaller but still sufficient for most tests.
Temperature range	Consumer / Industrial variants, up to **−40...+105 °C**	STM32H735 industrial grades (−40...+85/105 °C)	near	Both families have industrial-temperature parts.
Clock sources	24 MHz main crystal, 32.768 kHz RTC crystal, internal PLLs	25 MHz (or 8 MHz) main crystal, 32.768 kHz RTC, multiple PLLs	near	Both provide flexible PLL trees for system, USB, peripherals.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

2

Power Management

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
Power inputs	5 V via **barrel jack (J45)**, USB OTG (J47/J48), debug USB (J53)	5 V via **USB-C ST-LINK**, external VIN connectors	near	Both can be powered from USB and external 5 V sources.
Always-on / backup	Dedicated **SNVS** (secure non-volatile storage) domain with its own 3.3 V rail	Backup domain (VBAT) for RTC/backup registers	near	Both support small always-on domains; RT1060's SNVS is more elaborate but conceptually similar.
User power control	**ON/OFF button (SW8)** controlling PMIC_REQ_ON, **slide switch SW6**, POR SW7	Reset button; standard power-on via USB/external supply	not near	EVKC exposes full PMIC-style user control; H735 DK is simpler (reset only).
Regulators (on board)	External DC/DC + LDO tree (3.3 V, 1.8 V, SNVS_3V3, VDD_SOC_IN, etc.)	3.3 V external regulator + on-chip DCDC/LDO	near	Both use 3.3 V as main board rail then internal regulation to core/analog.

NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

3

Ethernet Support

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
Ethernet MAC	1× 10/100 Ethernet MAC (RMII) in i.MX RT1060	1× 10/100 Ethernet MAC (RMII) in STM32H735	near	Same class MAC; both support typical LwIP/network stacks.
PHY and connector	**KSZ8081RNB** PHY + RJ45 with integrated magnetics (J43)	On-board 10/100 PHY + RJ45 with magnetics (LAN8742A or similar)	near	Both boards provide a ready-to-use 10/100 RJ45 Ethernet port.

4

TrustZone / Security

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
TrustZone-M	**Not supported** (Armv7-M M7 core)	**Supported** (Armv8-M TZ-M)	not near	H735 can partition secure/non-secure; RT1060 cannot.
Other security IP	On-chip crypto, secure boot (through NXP security set)	Crypto accelerators, secure boot, etc.	near (partially)	Both support hardware crypto and secure boot, but only STM32 has TZ-M.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

5

Graphics & Display

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
LCD controller	**eLCDIF**, 24-bit RGB parallel interface	**LTDC**, 24-bit RGB parallel interface	near	Both have advanced dedicated LCD controllers with 24-bit RGB output.
On-board display	40-pin RGB connector (J49); **LCD not included** in basic kit, but recommended RK043FN66HS-CTG is 4.3" 480×272	**4.3" 480×272 TFT LCD with capacitive touch** mounted on the board	near	Same resolution and interface; STM board ships with the LCD, NXP board requires separate module.
Touch	Capacitive touch panel supported via connector (J49 B1–B6)	On-board capacitive touch for the LCD	near	Both support CTP; on H735 it is pre-wired.
2D graphics accel	Basic 2D (no GPU, uses eLCDIF and CPU for drawing)	Basic 2D (no GPU, uses LTDC and CPU/Chrom-ART if present on some SKUs)	near	Similar graphics model: software rendering + LCD controller.
Camera interface	**CSI parallel interface** + 24-pin FPC (J46); MT9M114 camera supplied	**No camera on H735G-DK** (no CSI routed on this board)	not near	RT1060-EVKC supports direct camera capture; H735G-DK doesn't.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

6

Timers & Control

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
General timers	Multiple 16/32-bit timers, motor-control capable PWM	Many advanced/general timers (16/32-bit) with PWM, encoder	near	Both provide sufficient timers for motor control, PWM, capture/compare tests.
Advanced timers	Capable motor/PWM timers but less specialized	Advanced motor-control timers (e.g. complementary PWM, dead-time)	near	For generic benchmarks both are adequate; STM32 may have richer motor peripherals but not essential for your use case.

7

Sensors

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
On-board sensors	**FXLS8974C** 3-axis accelerometer (I ² C, up to ±16 g)	MEMS sensors (typically accel, gyro, digital mic, depending on kit revision)	near	Both offer at least acceleration sensing; H735 DK may include additional environmental/MEMS sensors.

8

Debug & Tools

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
On-board debugger	**MCU-Link OB** (LPC55S69) – CMSIS-DAP or J-Link Lite, SWD+SWO, VCOM, can debug external target	**ST-LINK/V3E** – SWD+SWO, VCP (UART bridge), mass-storage drag-and-drop	near	Both provide a modern, integrated debug probe with SWD, SWO and VCOM.
External debug	20-pin JTAG/SWD (J2) & 5-pin SWD header for external targets	SWD via ST-LINK, plus standard SWD pads/headers	near	Both support attaching external debuggers/targets if needed.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

9

Middleware & Ecosystem

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
SDK / HAL	**MCUXpresso SDK** with HAL, drivers, middleware	**STM32CubeH7** with HAL/LL, drivers, middleware	near	Both vendors supply rich SDKs tailored to these boards.
RTOS support	FreeRTOS, bare-metal, others via MCUXpresso	FreeRTOS, Azure RTOS (ThreadX), bare-metal via Cube	near	Very similar RTOS ecosystems.
TCP/IP stack	LwIP in MCUXpresso SDK	LwIP in STM32CubeH7	near	Directly comparable Ethernet/IP middleware.
USB stacks	USB Device/Host (CDC, MSC, HID, audio, etc.)	USB Device/Host/OTG classes in Cube	near	Both have extensive USB stack support.
File system	FATFS on SD / external media	FATFS on SD / NOR / USB	near	Equivalent file-system middleware.
Graphics middleware	GUI libraries (e.g. emWin/LittlevGL integration in SDK examples)	TouchGFX / emWin / LVGL via STM32Cube	near	Both support modern embedded GUI frameworks targeting the LCD.
Audio middleware	I ² S audio examples (codec, USB audio, etc.)	I ² S / SAI audio + USB audio examples	near	Comparable audio stacks for playback/record, USB audio.



NXP: MIMXRT1060-EVKC Vs STM32 board – STM32H735G-DK

10

Connectivity (USB, SD, CAN,
Wi-Fi interface, expansion)

Criterion	NXP board – MIMXRT1060-EVKC	STM32H735G-DK	Tag	Reason
USB	**2x USB 2.0 OTG HS** controllers with integrated PHY; J48 (OTG1), J47 (OTG2)	USB HS (with PHY) + USB FS; typically, one micro-/USB-C port configured for OTG	near	Both can act as HS/FS device/host; port count/layout differs but capability comparable.
SD card	**microSD 3.0** on uSDHC1 (J22), 4-bit bus, 3.3 V/1.8 V selectable	**microSD slot** on SDMMC interface	near	Both support high-speed SD card with 4-bit bus for file systems.
CAN	FlexCAN + **TJA1057GT** transceiver, CAN connector footprint (J42)	CAN FD controller + on-board transceiver (varies by rev, typically present)	near	Both enable immediate CAN bus experiments; STM32 has CAN FD capability.
Wi-Fi/BLE interface	**M.2 (Key-E) slot** with SDIO + UART + I ² S + I ² C, LEDs for Wi-Fi/BLE	No native M.2; Wi-Fi/BLE via external STMod+/Arduino shields	not near	RT1060-EVKC has a dedicated M.2 wireless socket; STM board relies on generic expansion.
Other expansion	**Arduino-compatible headers**, **mikroBUS socket**, I ² C header, radio header	**Arduino Uno-style headers**, **STMod+** connectors	near	Both boards integrate into popular shield ecosystems (Arduino-style + vendor-specific bus).



NXP board – FRDM-MCXA156

The FRDM-MCXA156 board is a design and evaluation platform based on the NXP MCXA156 microcontroller (MCU). The MCXA156 MCU is a low-power microcontroller for industrial and consumer Internet of Things (IoT) applications. It has one Arm Cortex-M33 core running at speeds of up to 96 MHz. It supports industrial communication protocol, brushless direct current (BLDC) motor / permanent magnet synchronous motor (PMSM) control, and integrated sensor interfaces (MIPI I3C, I2C, and SPI).



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

1

Core & Clock System

Criterion	NXP board – FRDM-MCXA156	STM32 board – NUCLEO-L552ZE-Q	Tag	Reason
Core type	Arm **Cortex-M33**	Arm **Cortex-M33**	near	Same Armv8-M core; ideal for architectural comparison.
Max core frequency	Up to **96 MHz**	Up to **110 MHz**	near	Both are mid-frequency low-power M33 cores; STM32 slightly higher.
Internal code Flash	MCXA156 family: typically **512 KB** class (device-dependent)	512 KB dual-bank Flash (STM32L552ZET6Q)	near	Same Flash order of magnitude; suitable for similar application sizes.
Internal SRAM	MCXA156 family: typically **256 KB** class	256 KB SRAM	near	Similar SRAM size; good for comparable RTOS/middleware usage.
External memory IF	None (no SDRAM/OSPI routed on FRDM-MCXA156)	None populated; STM32L552 does not target external RAM	near	Both boards are internal-memory-centric, simplifying fair comparison.
Package / pins	MCXA156VLL – **100-pin LQFP**	STM32L552ZET6Q – **144-pin LQFP**	not near	STM32L5 device has more pins and potential I/O than MCXA156.
Temperature range	–40...105 °C (industrial variants)	–40...85/105 °C (L552 series)	near	Both families offer industrial-temperature options.
Clock sources	8 MHz main crystal for MCXA156; 16 MHz for debug MCU; CLKOUT test point	MSI/HSI internal RCs; optional HSE; 32.768 kHz LSE/LSI; PLL	near	Both provide internal RCs and external crystal options for precise timing and LP operation.



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

2

Power Management

Criterion	NXP board – FRDM-MCXA156	STM32 board – NUCLEO-L552ZE-Q	Tag	Reason
Power inputs	5 V via USB-C **J23** (MCU USB FS) or USB-C **J21** (MCU-Link); 5–9 V via Arduino VIN	5 V via ST-LINK USB, external VIN via Arduino power header	near	Both can be powered from USB or external 5 V/5–9 V sources.
Main 3.3 V rail	LDO **U2 (XC6227C331)** → LDO_3V3; feeds MCU, MCU-Link, CAN, sockets, LCD/camera	3.3 V rail from on-board LDO/DC-DC feeding MCU and I/O	near	Similar architecture: 5 V in, regulated 3.3 V domain.
Current measurement	JP1/JP2/JP3/JP4 jumpers allow separate measurement of board, digital, analog, total MCU	Nucleo allows breaking VDD link to measure target current	near	Both platforms support explicit current measurement for power benchmarks.
Backup / RTC supply	VBATT not separately broken out (RTC powered from main 3.3 V)	Dedicated VBAT pin; some Nucleo revisions allow backup battery	near	Both can support RTC backup, STM32 has explicit VBAT pin.



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

3

USB Support

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
USB role (target)	USBFS0: **USB 2.0 Full Speed Device** via Type-C J23	USB FS device via STM32L552 (Micro-B on Nucleo)	near	Both provide USB FS device, suitable for CDC/MSC/HID middleware.
USB Host / OTG	Not supported by USBFS0; device-only	STM32L5 FS is device-oriented; host capability limited on this board	near	Both are primarily FS device-mode; no HS host complexity.
Debug USB	MCU-Link via Type-C J21 (debug + VCOM)	ST-LINK USB (debug + VCP + mass-storage)	near	Separate debug USB does not interfere with application USB.

4

Industrial Buses (CAN, I³C)

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
CAN / CAN FD	FlexCAN0 with **CAN FD** and TJA1057 FD transceiver on J22	No CAN interface on STM32L552 Nucleo	not near	CAN-based middleware (ISO-TP, CANopen) only directly testable on FRDM.
I ³ C	**I3C0** controller used with P3T1755 sensor	No I ³ C	not near	Any I ³ C middleware / driver comparison is unique to FRDM-MCXA156.



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

5 UART/SPI/I²C

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
UART	5× LPUART; routed to MCU-Link VCOM, Arduino, mikroBUS	Several USART/LPUARTs routed to ST-LINK VCP and headers	near	Both support serial console and protocol stacks (Modbus, SLIP, etc.).
SPI	2× LPSP1 (mikroBUS/Pmod + Arduino)	Multiple SPI on Arduino/morpho	near	Suitable for SPI-based middleware (SD card, external flash).
I ² C	4× LPI2C; multiple busses to Arduino, LCD, camera, mikroBUS/Pmod	3× I ² C including LP I ² C; exposed on Arduino/morpho	near	Comparable for sensor-stack middleware and I ² C-based components.

6 ADC / Timers

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
ADCs	2× 12-bit ADCs; channels routed to Arduino & mikroBUS; motor signals	2× 12-bit ADCs with oversampling	near	Both suitable for RTOS + ADC driver tests (e.g. periodic sampling, DMA).
Timers / SysTick	PWM0, PWM1, general timers, SysTick	General + low-power timers, advanced PWM, SysTick	near	Both can drive RTOS ticks, timebase and periodic tasks.

7 Expansion Ecosystem

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
Arduino UNO	Full Arduino UNO + A4/A5 compatible sockets	Arduino UNO R3-compatible headers	near	Can run identical shield-based middleware demos (e.g. SD over SPI).
mikroBUS / Pmod	**On-board mikroBUS and Pmod footprint**	No native mikroBUS/Pmod	not near	FRDM board integrates more expansion standards without extra hardware.



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

8

Sensors & HMI

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
On-board sensor	**P3T1755 digital temperature sensor (I ² C/I ³ C)**	No discrete sensor (MCU internal only)	not near	FRDM can run sensor-focused middleware/RTOS examples without extra hardware.
LEDs & buttons	Power LED, reset LED, RGB user LED; reset, wake-up, ISP buttons	User and reset buttons, user LED	near	Both adequate for RTOS + UI events, though FRDM has more diagnostic LEDs.

9

Debug & Tooling

Criterion	FRDM-MCXA156	NUCLEO-L552ZE-Q	Tag	Reason
On-board debugger	MCU-Link OB (LPC55S69 M33) – CMSIS-DAP or J-Link Lite, SWD/SWO, VCOM	ST-LINK/V2-1 or V3 – SWD/SWO, VCP, MSD	near	Both support RTOS-aware debug, SWO traces, and UART logging.
External debug	2×5 SWD header J24	SWD header pads or connector	near	Both can use third-party probes for advanced RTOS trace tools.



NXP board – FRDM-MCXA156 Vs STM32 board – NUCLEO-L552ZE-Q

10

Middlewares

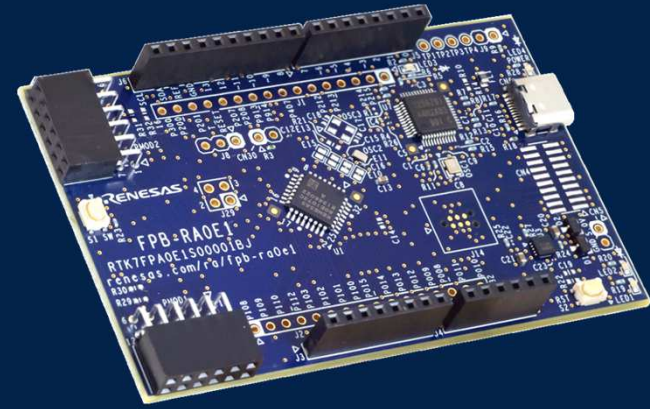
Criterion	FRDM-MCXA156 (NXP MCXA156)	NUCLEO-L552ZE-Q (STM32L552)	Tag	Reason
RTOS options	**MCUXpresso SDK** fully supports FreeRTOS; examples for tasks, queues, timers; CMSIS-RTOS2 supported via SDK	**STM32CubeL5** integrates FreeRTOS and CMSIS-RTOS2; additional Azure RTOS (ThreadX) support	near	Both ecosystems provide mature RTOS options and integration with their HAL.
RTOS-aware debug	MCUXpresso IDE with RTOS-aware views for FreeRTOS; SWO trace via MCU-Link	STM32CubeIDE + ST-LINK RTOS-aware debug for FreeRTOS; SWO trace	near	RTOS instrumentation available on both.
File system support	**FatFS** integrated in MCUXpresso SDK (SD card, SPI flash); examples for block devices	**FatFS** integrated in STM32CubeL5 (SD/MMC, NOR/NAND, USB MSC media)	near	Both platforms support the same de-facto standard embedded FS stack.
USB Device middleware	USB FS device stack in SDK: CDC, HID, MSC, Audio, etc.; examples for FRDM boards	USB Device stack in CubeL5: CDC, HID, MSC, Audio, DFU, etc.	near	Very comparable coverage of USB device classes.
USB Host / OTG middleware	RTOS+USB host less relevant (MCXA156 is device-only on this board)	STM32L552 FS is mainly device; host examples limited on this Nucleo	near	Both are primarily device-centric; host tests are out of primary scope.
Networking stacks	LwIP and higher-level stacks generally available for NXP M33 family, but MCXA156 FRDM board **has CAN FD and I ³ C**, not Ethernet	STM32L5 family has LwIP through Cube; Nucleo-L552ZE-Q **has no Ethernet or CAN hardware on board**	not near	Ethernet-based TCP/IP benchmarks are better on other boards; CAN-based stacks only on FRDM.
Security / TrustZone middleware	MCUXpresso SDK + MCXA156 support secure/non-secure firmware split and secure boot (documentation in RA/MCX security manuals)	STM32CubeL5 provides **TrustZone + TF-M** examples, secure services, and SBSFU secure bootloader	near	Both platforms can be used to compare secure RTOS and secure services, though exact frameworks differ.
Motor-control libraries	NXP provides examples and motor-control middleware for FRDM + FRDM-MC-LVBLDC/LVPMSM boards	STM32L5 is not the primary motor-control line; ST motor libraries target G4/F3/H7 families	not near	FRDM-MCXA156 is more suitable for motor-control middleware tests than Nucleo-L552.



Renesas FPB-RA0E1

The FPB-RA0E1, a Fast Prototyping Board for the RA0E1 MCU Group, enables users to seamlessly evaluate the features of the RA0E1 MCU group and develop embedded systems applications using Flexible Software Package (FSP) and the e2 studio IDE. Users can use on-board features along with their choice of popular ecosystems add-ons to bring their big ideas to life.

The key features of the FPB-RA0E1 board are categorized in two groups (consistent with the architecture of the board) as follows:



Renesas FPB-RA0E1 vs NUCLEO-L031K6

1

Core & Memory

Criterion	Renesas board – FPB-RA0E1 (RA0E1)	STM32 board – NUCLEO-L031K6 (STM32L031K6)	Tag	Reason
Core type	Arm **Cortex-M23**	Arm **Cortex-M0+**	near	Both are small, low-cost Arm cores; M23 is Armv8-M baseline, M0+ is Armv6-M, same application class.
Max core frequency	**32 MHz**	**32 MHz**	near	Identical nominal clock; timing results can be compared directly or normalized easily.
Internal code Flash	**64 KB**	32 KB	near	Same order of magnitude; RA0E1 has more Flash but still "small-memory MCU" class.
Internal SRAM	**12 KB**	8 KB	near	Both have very limited RAM; RA0E1 gives slightly more headroom.
Data Flash / EEPROM	1 KB Data Flash	No dedicated data Flash	not near	RA0E1 includes a small NVM region useful for parameters; L031 emulates in main Flash.
External memory IF	None	None	near	Neither supports SDRAM/OSPI; all code/data in internal memory.
Package / pins	32-pin LQFP; all pins can be accessed via J1/J2 headers	32-pin LQFP; all pins exposed on Nano-style headers	near	Same pin count and footprint scale.



Renesas FPB-RA0E1 vs NUCLEO-L031K6

2

Power & Current Measurement

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
Power input	5 V from debug USB-C (CN6) or external 5 V on CN5 test header	5 V from ST-LINK micro-USB or external VIN on header	near	Both use 5 V as main input and regulate down to 3.3 V.
3.3 V rail	Single 5→3.3 V LDO (2 A) powering RA0E1 + connectors	5→3.3 V regulator powering STM32 + connectors	near	Similar simple power tree.
Current measurement	0 Ω **R3** between VCC and VCC_MCU; remove R3 and use **CN30** for ammeter/shunt	VDD-target link or dedicated jumper can be opened to insert ammeter	near	Both boards support cutting the MCU VDD path to measure current.
Power indication	LED4 (green) power; LED3 (yellow) debug	Power LED + ST-LINK status LED	near	Clear visual power/debug states on both.



Renesas FPB-RA0E1 vs NUCLEO-L031K6

3

Debug & VCOM

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
On-board debugger	**RA4M2 with SEGGER J-Link OB**, SWD via CN6, SWDIO on P108, SWCLK on P300	**ST-LINK/V2-1**, SWD via micro-USB	near	Both provide integrated SWD probes, no external tools required for basic debug.
External debug	10-pin Cortex SWD header **CN4**	6/10-pin SWD header/patch on Nucleo	near	Both can use external SWD probes if needed.
VCOM / UART bridge	J-Link CDC UART mapped to RA0E1 TXDA0/RXDA0 (P208/P207) when E4/R10 configured	ST-LINK VCP mapped to an STM32 USART	near	Both offer USB-UART bridging for logs/CLI.

4

Clocks

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
Internal clocks	Multiple on-chip high/medium/low-speed RC oscillators	32 MHz HSI RC, 37 kHz LSI	near	Both rely primarily on internal RC oscillators for MCLK.
External clocks	32.768 kHz sub-clock crystal fitted; optional 20 MHz main crystal via OSC3 and E13/E15	Optional HSE/LSE crystals on Nucleo (board-dependent)	near	Both can add external crystals where precision timing or RTC accuracy is needed.

5

Digital I/O, UART, SPI, I²C

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
UART	SAU channels; TXDA0/RXDA0 on Arduino D0/D1; Pmods configurable as UART	Up to 2x USART/UART; one connected to ST-LINK VCP and header pins	near	Both boards support at least one application UART plus a debug UART.
SPI	SAU “Simple SPI” on Pmod1/Pmod2; SPI-capable Arduino pins D10–D13	SPI (SPI1) exposed to Nano headers	near	Both allow simple SPI communication to shields and sensors.
I ² C	SAU can emulate I ² C; Pmod Type-6A uses I ² C; Arduino A4/A5 used for SDA/SCL	I ² C peripheral mapped to pins; SDA/SCL on Nano headers	near	Suitable for I ² C-based sensor stacks on both boards.



Renesas FPB-RA0E1 vs NUCLEO-L031K6

6

Ecosystem & Expansion

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
Arduino connectors	**Full Arduino Uno R3** headers (J3–J6: power, analog A0–A5, D0–D13, SDA/SCL)	No UNO footprint; Nano-style Nucleo dual headers	not near	RA0E1 natively supports Uno shields; L031K6 targets "Nano" form factor.
Pmod connectors	**Two 12-pin Pmod headers** (Pmod1, Pmod2; Type 2A/3A/6A via jumpers)	No native Pmod; needs adapter board	not near	RA0E1 integrates Pmod; Nucleo requires extra hardware.
Breadboard access	J1/J2 breakout (not fitted) provide all MCU pins and power on 2.54 mm grid	Nucleo headers expose all pins in 2.54 mm grid	near	Both can be plugged into breadboards or wired directly for prototypes.

7

Analog & Timers

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
ADC	12-bit ADC; inputs on Arduino A0–A5 and other pins	12-bit ADC; several channels exposed on headers	near	Both support basic analog measurement with similar resolution.
DAC	None	None on L031K6	near	No DAC comparison required.
Timers/PWM	Timer/PWM channels on Arduino D3/D5/D6/D9–D11 and Pmods	Basic/general timers with PWM on Nano header pins	near	Both can generate PWM outputs needed for motor/sensor control demos.

8

LEDs & Buttons

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
User LEDs	LED1 (green, P008), LED2 (green, P009)	Typically, 1 × user LED (LD2)	near	Both provide visible user feedback; RA0E1 has more LEDs for richer UI.
Debug/power LED	LED3 (debug status) + LED4 (power, green)	ST-LINK and power LEDs	near	Comparable debugging and power indication.
User button	S1 user switch on P200/NMI/IRQ0	User button on a GPIO	near	One user button on each for input tests.
Reset button	S2 reset to RES/P206	Reset button tied to NRST	near	Similar manual reset functionality.



Renesas FPB-RA0E1 vs NUCLEO-L031K6

9

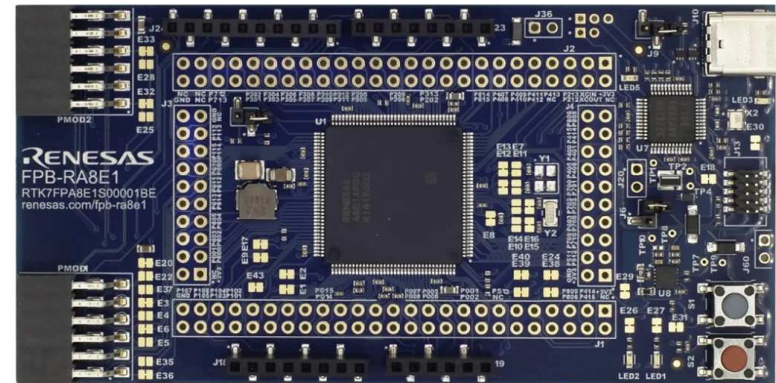
Middleware & Software

Criterion	Renesas board – FPB-RA0E1	STM32 board – NUCLEO-L031K6	Tag	Reason
Vendor SDK	**Renesas FSP** with RA0E1 BSP, drivers, and examples	**STM32CubeL0** with HAL/LL, BSP, and examples	near	Both are well covered by vendor SDKs tailored to these boards.
RTOS usage	FreeRTOS supported via FSP; fits but tight at 64 KB/12 KB	FreeRTOS supported via CubeL0; tighter at 32 KB/8 KB	near	Both can run small RTOS demos; RA0E1 has slightly more memory margin.
USB application	RA0E1 has **no USB peripheral**; only J-Link VCOM is USB-based	STM32L031 has no USB; only ST-LINK VCP is USB-based	near	No native USB device/host on either; only debug USB; middleware USB stacks not primary focus.
File system	Can use FatFS with external SPI Flash/SD shield via Arduino/Pmod	Can use FatFS with external media via SPI on shields	near	FS tests require external media on both; stacks available.
Example coverage	GPIO/UART/SPI/I ² C/ADC/low-power examples in FSP	Similar peripheral and low-power examples in CubeL0	near	Comparable software starting points for low-end control applications.



Renesas FPB-RA8E

The FPB-RA8E1, a Fast-Prototyping Board for the RA8E1 MCU Group, enables users to seamlessly evaluate the features of the RA8E1 MCU group and develop embedded systems applications using Flexible Software Package (FSP) and the e2 studio IDE. Users can use on-board features along with their choice of popular ecosystems add-ons to bring their big ideas to life. The key features of the FPB-RA8E1 board are categorized in two groups (consistent with the architecture of the board) as follows:



FPB-RA8E1 vs NUCLEO-H743ZI2

1

Core & Memory

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
Core type	Arm **Cortex-M85**	Arm **Cortex-M7**	near	Both are high-end cores intended for intensive embedded processing; M85 is newer but close in class to M7.
Max core frequency	**360 MHz**	**400 MHz**	near	Same frequency class; Nucleo runs slightly faster.
Internal Flash	**1 MB** code Flash + 12 KB Data Flash	**2 MB** Flash	near	Both have large Flash; STM32 has more capacity but same order of magnitude.
Internal SRAM	**544 KB** SRAM including 32 KB TCM	**1 MB** SRAM (AXI SRAM + ITCM/DTCM)	near	Both offer plenty of on-chip SRAM, with TCM regions; STM32 has larger total size.
External memory IF	None populated; no SDRAM/OSPI on board	FMC + QUADSPI available in MCU, but no external memory fitted on Nucleo	near	Both boards are internal-memory-centric in their default configuration.
Package / pins	RA8E1AFDCFB – **144-pin LQFP**	STM32H743ZI – **144-pin LQFP**	near	Same pin count/package; similar I/O density.
Temperature range	Industrial range (–40...105 °C) for RA8E1 family	–40...85/105 °C (STM32H743 industrial grades)	near	Both families target industrial/embedded temperature ranges.



FPB-RA8E1 vs NUCLEO-H743ZI2

2

Power & Current Measurement

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
Power inputs	5 V via debug USB-C **J10**, or 5 V via header **J60**, or 5 V via test points TP7/TP9	5 V via ST-LINK USB; external 5 V/VIN via Arduino power pins	near	Both accept 5 V from USB and external connectors.
Main 3.3 V rail	5 V → 3.3 V on-board LDO; 3.3 V powers MCU and peripherals	5 V → 3.3 V regulator/DC-DC powering STM32 and board	near	Similar power-tree concept.
RTC backup	VBATT pin on RA8E1; battery via **J36** (not fitted)	VBAT pin on STM32H7 + optional battery header on some boards	near	Both support RTC/backup domain via a backup supply.
Current measurement	Precision **5 mΩ R2** in MCU 3.3 V path; measure voltage across TP2/TP4 or remove R2 and use J20	VDD-target link on Nucleo can be cut and measured	near	Both provide explicit mechanisms to measure MCU current consumption.

3

Debug & Trace

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
On-board debugger	**RA4M2 + SEGGER J-Link OB**, SWD via USB-C J10	**ST-LINK/V3E**, SWD/SWO via micro-/USB-C (board rev-dependent)	near	Both offer integrated high-performance debug probes (SWD, SWO, USB).
Debug interfaces	SWD (on-board) + 10-pin JTAG/SWD header **J13** for SWD/JTAG/SWO	SWD via ST-LINK; 20-pin STDC14 or SWD header pads	near	External trace/debug tools can be connected on both boards.
Boot/Debug jumpers	J6/J9 for boot/debug mode selection (Debug on-board vs Debug-in)	BOOT0 & CN2/boot jumpers for boot selection	near	Both have boot/debug jumpers, though implementation differs.



FPB-RA8E1 vs NUCLEO-H743ZI2

4

Ecosystem & Expansion

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
Arduino UNO support	**Arduino UNO R3** compatible connectors (J11, J14, J15, J18)	**Arduino UNO R3** compatible headers + ST morpho headers	near	Both can accept standard Arduino shields and extensions.
Pmod connectors	**2× 12-pin Pmods**: Pmod1 (SPI/I ² C/UART), Pmod2 (SPI/UART)	No native Pmod connectors	not near	RA8E1 has on-board Pmod; Nucleo needs adapter boards for Pmod modules.
Breakout headers	J1, J2, J3 (not fitted) expose all RA8E1 I/O and power pins	Morpho headers on both sides expose all STM32H7 pins	near	Both provide full access to MCU pins for custom mezzanines/breadboards.
Camera expansion	**20-pin camera connector J4** for ArduCam OV3640 (parallel RGB/YUV/JPEG)	No camera connector on Nucleo-H743ZI2	not near	FPB-RA8E1 is camera-ready; H743 Nucleo requires extra hardware for camera.

5

USB & Communication

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
USB on MCU	RA8E1 device USB (board text hints at device USB ports; actual connector for device not on base FPB, focus is debug USB)	H743 has USB HS/FS; on Nucleo mainly exposed via micro-B (FS)	not near	Nucleo-H743 has a more complete USB FS/HS device capability exposed; FPB-RA8E1 focuses on debug USB only.
UART	SCI peripherals, e.g. UART3 on Arduino D0/D1 (P309/P310); further UARTs on Pmods	Multiple USARTs/ UARTs on Arduino & morpho; one linked to ST-LINK VCP	near	Both have multiple UARTs for console and interfaces.
SPI	Several SCI channels in “Simple SPI” mode; Arduino SPI pins D10–D13; Pmod1 and Pmod2 for SPI	Several SPI peripherals routed to Arduino & morpho headers	near	Both can drive SPI-based peripherals and expansion boards.
I ² C	I ² C bus via SDA1/SCL1 on Arduino (P511/P512) and on Pmod1; also to camera header	Multiple I ² C buses on H743 with pins on Arduino/morpho	near	I ² C-based sensor/display stacks can be realized on both.



FPB-RA8E1 vs NUCLEO-H743ZI2

6

Analog & Timers

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
ADC	RA8E1: 12-bit ADCs; analog inputs A0–A5 on Arduino; general-purpose analog	STM32H743: multiple 12-/16-bit ADCs with many channels on morpho	near	Both support multi-channel 12-bit ADC; STM32 has more channels and bandwidth but same class.
DAC	RA8E1 has at least one DAC channel (P014/DA0 on Arduino A4)	H743 includes 2-channel 12-bit DAC	near	Both provide DAC outputs; STM32 has more but same functional capabilities.
Timers / PWM	GTIOC advanced timers used on Arduino D2–D7, D9–D13 for PWM (GTIOC0A/2A/2B/3A/3B/4A/4B/5A/5B)	Large set of 16-/32-bit timers; advanced timers TIM1/TIM8 for PWM and motor control	near	Both boards expose multiple PWM outputs for motor control and timing benchmarks.

7

User I/O (LEDs, Buttons)

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
User LEDs	LED1 & LED2 (green) on P404, P408; LED3 (power); LED5 (debug)	LD2 user LED + power & ST-LINK LEDs	near	Sufficient user/status LEDs on both boards.
User button	S1 on P009 (IRQ13-DS)	User button (B1) on a GPIO	near	One user button on each for wakeup or input.
Reset button	S2 connected to RESET#	Reset button on NRST	near	Both allow manual reset.

8

Camera & Graphics

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
Camera support	**Camera expansion via J4** (ArduCam OV3640 3 MP parallel RGB/YUV/JPEG)	No on-board camera interface; requires external board	not near	FPB-RA8E1 is much better suited for camera/vision development.
LCD / graphics	No dedicated LCD controller on FPB-RA8E1 board (graphics done via GPIO or external display through SPI/I ² C)	H743 has LTDC for RGB LCD, but Nucleo board does not include an LCD connector by default	not near	For out-of-the-box graphics, neither board has LCD; RA8E1 has camera, H743 has LTDC in silicon but not wired.



FPB-RA8E1 vs NUCLEO-H743ZI2

9

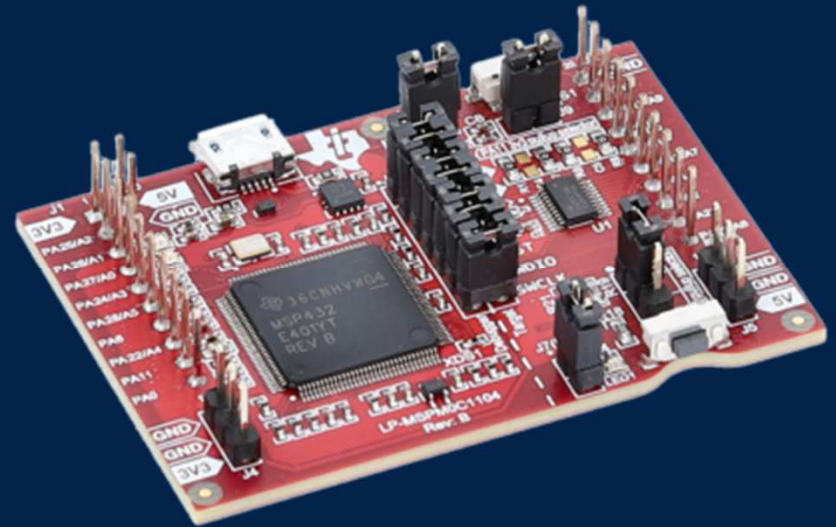
Middleware & Software

Criterion	Renesas board – FPB-RA8E1	STM32 board – NUCLEO-H743ZI2	Tag	Reason
SDK / HAL	**Renesas FSP** with RA8E1 BSP, drivers, FreeRTOS examples, security examples	**STM32CubeH7** with HAL/LL, FreeRTOS, security examples	near	Both are flagship boards within their vendor ecosystems with rich SDK support.
RTOS support	FreeRTOS via FSP; RTOS-aware debug in e ² studio via J-Link	FreeRTOS & Azure RTOS via Cube; RTOS-aware debug in CubeIDE	near	RTOS features and tooling are comparable.
File system	FatFS via FSP on external storage (e.g. SPI Flash/SD on shields)	FatFS via CubeH7 on SD/NOR/USB MSC media (external)	near	Both use FatFS for file systems with similar APIs and example projects.
USB middleware	RA8E1 device stacks via FSP (CDC/HID/MSC etc.), though FPB board does not bring a separate device port out by default	H743 FS/HS device & host stacks (CDC/HID/MSC/DFU, etc.); Nucleo provides physical FS connector	near*	In silicon, both support rich USB stacks; H743 Nucleo is better wired for direct USB device tests.
Networking (Ethernet)	No Ethernet MAC/PHY on RA8E1 device or FPB-RA8E1 board	H743 MCU includes Ethernet MAC, but Nucleo board lacks PHY/RJ45	not near	Neither board is ready for Ethernet benchmarks; H743 needs additional hardware.
Camera-related SW	FSP + examples for camera expansion (OV3640 + RA8E1)	No official camera example for this Nucleo by default	not near	FPB-RA8E1 has direct camera middleware paths; Nucleo-H743ZI2 does not.



TI LP-MSPM0C1104

The MSPM0C1104 LaunchPad™ Development Kit is an easy-to-use evaluation module for the MSPM0C1104 microcontroller (MCU). The kit contains everything needed to start developing on the MSPM0C110x microcontroller platform, including onboard debug probe for programming and debugging. The board also features an onboard button and LED for quick integration of a simple user interface.



LP-MSPM0C1104 vs NUCLEO-L031K6

1

Core & Memory

Criterion	TI board – LP-MSPM0C1104 (MSPM0C1104)	STM32 board – NUCLEO-L031K6 (STM32L031K6)	Tag	Reason
Core type	Arm **Cortex-M0+**	Arm **Cortex-M0+**	near	Same core architecture; both are small, low-end MCUs.
Max core frequency	**24 MHz**	**32 MHz**	near	Same order of magnitude; STM32 runs slightly faster.
Code Flash	**16 KB**	32 KB	near	Both are “small-Flash” devices; STM32 has 2× Flash but still in same segment.
SRAM	**1 KB**	8 KB	not near	STM32 has much more RAM; MSPM0C1104 is significantly more constrained.
Data NVM	None separate from code Flash	None (no dedicated EEPROM/Data Flash)	near	Both rely on main Flash for non-volatile data storage.
External memory IF	None	None	near	Neither supports external SDRAM/OSPI; pure internal-memory MCUs.



LP-MSPM0C1104 vs NUCLEO-L031K6

2 Power & Current Measurement

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
Power input	5 V via USB micro-B (debug); or external 5 V/3.3 V via BoosterPack/power headers	5 V via ST-LINK USB and external VIN pins	near	Both take 5 V from USB and can be powered from external rails.
3.3 V rail	5 V → 3.3 V LDO (TPS73533) up to 500 mA	5 V → 3.3 V regulator powering MCU and headers	near	Simple single-rail supplies on each board.
Current measurement	Remove **3V3** jumper on isolation block J101 and insert ammeter	Break VDD-target link (user jumper) and insert ammeter	near	Both allow inserting a meter in the VDD path to measure target current.

3 Debug & VCOM

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
On-board debugger	**XDS110** (MSP432E401Y), SWD & backchannel UART	**ST-LINK/V2-1** ¹ , SWD & VCP (virtual COM port)	near	Both provide integrated SWD debug probes and USB-UART bridges.
Isolation	J101 isolation block separates XDS110 domain from target (5 V, 3.3 V, SWD, UART)	ST-LINK can be isolated via VDD link; SWD pins accessible	near	Both allow you to fully disconnect debugger for power measurements or use the probe externally.
External debug	XDS110 can be repurposed to debug external devices via J101	ST-LINK can be used as standalone probe on many Nucleos	near	Both probes can target other boards if desired.

4 Clocks

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
Internal clocks	SYSOSC 24 MHz (2.5%); MCLK defaults from 32 MHz SYSOSC; ULPCCLK derived from MCLK	HSI 32 MHz internal RC; LSI 37 kHz	near	Both primarily use internal RC oscillators with similar accuracy.
External clocks	No crystal required on this LaunchPad (internal only)	Optional LSE (32.768 kHz) and HSE crystals (board-dependent)	near	External crystals can be fitted on STM32; MSPM0 relies on internal RC in this kit.



LP-MSPM0C1104 vs NUCLEO-L031K6

5 Expansion / Ecosystem

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
Primary expansion	**20-pin BoosterPack headers** (LaunchPad standard)	**Nano-style dual headers** exposing all pins	near	Both expose MCU pins with 2.54 mm headers; ecosystems differ (BoosterPack vs Nucleo-Nano).
Ecosystem match	TI BoosterPacks: wireless, displays, sensors, power modules	ST shields / custom Nano boards	not near	Physical/form-factor ecosystems differ, though conceptually similar.
RC filter for ADC/PWM	**On-board RC networks** on selected pins for ADC input or PWM “DAC”	No dedicated RC filters; user adds externally	not near	LP-MSPM0C1104 includes RC filtering by design.

6 Digital I/O & Serial Interfaces

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
UART	UART0 routed to BoosterPack pins and to backchannel UART via XDS110	USART/UARTs available on headers; one mapped to ST-LINK VCP	near	Both have at least one application UART and one debug UART.
SPI	SPI0 (SPI_SCK, POCI, PICO, CSx) on BoosterPack pins	SPI1 on Nano headers	near	Both boards support basic SPI peripherals for external devices.
I ² C	I ² C0 (SCL/SDA) to BoosterPack pins	I ² C bus on Nano headers	near	Both provide at least one I ² C bus for sensors and peripherals.

7 Analog & Timers

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
ADC	**12-bit SAR ADC**, 866 ksp/s, 10 channels, with RC-filtered inputs	12-bit ADC (up to ~1 Msps) on several analog pins	near	Both are capable 12-bit ADCs; MSPM0 emphasises speed, STM32 emphasises flexibility.
DAC	None	None (L031K6)	near	No DAC in either device.
Timers/PWM	TIMA/TIMG timers, PWM outputs on various BoosterPack pins	General-purpose timers; PWM on Nano headers	near	Both are adequate for simple timing and PWM workloads.



LP-MSPM0C1104 vs NUCLEO-L031K6

8

Middleware

Criterion	TI board – LP-MSPM0C1104	STM32 board – NUCLEO-L031K6	Tag	Reason
Vendor SDK	**MSPM0 SDK** (driverlib, 50+ examples, TI Clang support)	**STM32CubeL0** (HAL/LL, BSP, examples)	near	Both boards have mature vendor SDKs with examples for GPIO, ADC, UART, timers, etc.
RTOS suitability	FreeRTOS possible but tight in **16 KB/1 KB**; often used bare-metal	FreeRTOS technically possible but also tight in 32 KB/8 KB; many examples are bare-metal	near	Both can run minimal RTOS setups but are primarily aimed at bare-metal control.
USB device/host	MSPM0C1104 has **no USB controller**; only XDS110-side USB for debug/VCOM	STM32L031K6 also has **no USB**; only ST-LINK VCP	near	No native USB stack on either MCU; comparison focuses on debug USB only.
File system	Via external SPI/SD BoosterPack if desired; MSPM0 SDK includes FatFS support on larger devices	Via external SPI/SD shield; CubeL0 includes FatFS middleware	near	Both can demonstrate FS with external media; not typical for these small MCUs.
Example coverage	Out-of-box GUI, LED toggle, GPIO, ADC, UART, timer examples via MSPM0 SDK	GPIO, ADC, UART, timer, low-power examples via CubeL0	near	Similar level of starter software for entry-level development.



GD32E5072Z-EVAL

The GD32E507Z-EVAL board is a hardware platform that enables prototyping on GD32E507ZE Cortex-M33 High Performance MCU.

The GD32E507ZE features a single-core ARM Cortex-M33 MCU which can run up to 180 MHz with flash accesses zero wait states, 512kiB of Flash, 128kiB of SRAM and 112 GPIOs.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

1

Core & Memory

Criterion	GD32E507ZE (GD32E507Z-EVAL)	STM32F429ZI (NUCLEO-F429ZI)	Tag	Reason
Core type	Arm Cortex-M33 , ARMv8-M, FPU + DSP	Arm Cortex-M4F , ARMv7-M, FPU + DSP	not near	Different core generation; both high-performance, but M33 vs M4 affects instruction set (TrustZone support on M33, different security model).
Max core frequency	180 MHz	180 MHz	near	Same maximum CPU frequency; similar performance class.
Flash	Up to 512 KB	2 MB	not near	STM32 has 4× more internal Flash, suitable for larger stacks/middleware.
SRAM	Up to 128 KB	256 KB (plus small backup SRAM)	near	STM32 has 2× RAM, but both are “hundreds of KB” class; same order of magnitude.
External memory IF	EXMC (SRAM, PSRAM, NOR, NAND, PC Card), SQPI	FMC (SRAM, PSRAM, NOR, NAND, SDRAM) + Quad-SPI	near	Both support code/data from external parallel + serial memories; exact feature set differs (SDRAM vs SQPI), but capabilities are comparable.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

2

Power & Clocks

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
VDD range	**1.71–3.63 V**	**1.8–3.6 V**	near	Overlapping range; both support typical 3.3 V systems and low-voltage operation.
Low-power modes	Sleep, Deep-sleep, Deep-sleep1/2, Standby	Sleep, Stop (0/1), Standby	near	Both have multi-level low-power modes; names differ but concept similar.
High-speed clocks	IRC8M, HXTAL 4–32 MHz, complex PLL tree	16 MHz HSI, HSE 4–26 MHz, PLLs up to 180 MHz	near	Both rely on internal RC + external crystal + PLL for 180 MHz system clock.
USB clock support	Dedicated IRC48M + PLLUSB up to 480 MHz	Dedicated PLLQ for 48 MHz USB clock	near	Each has special PLL resources for USB clocks.

3

Analog & Mixed-Signal

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
ADC	**2 × 12-bit**, up to **2.5 MSPS**, 16 external channels	**3 × 12-bit**, up to ~**2.4 MSPS**, many channels	near	ADC resolution and speed are very close; STM32 offers one extra ADC and more channels.
ADC oversampling	Hardware oversampling up to effective 16-bit	Hardware oversampling (series-dependent)	near	Both can improve SNR/resolution via oversampling.
DAC	**1 × DAC**, **2 channels**, 12-bit	**1 × DAC**, **2 channels**, 12-bit	near	Very similar DAC capabilities (2ch 12-bit).
Comparators	**3 × comparators**	No hard comparators in F429 core line (only analog threshold functions via other means)	not near	GD32 integrates explicit CMP IP; F429 relies on external comparators or OpAmps (not built-in).
Temp sensor	Yes (internal, via ADC)	Yes (internal, via ADC)	near	Both provide internal temperature sensing.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

4

Timers & Motor / Power Control

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
Advanced PWM timers	**2 × 16-bit advanced** (TIMER0, TIMER7)	**2 × 16-bit advanced** (TIM1, TIM8)	near	Both can drive 3-phase motors with complementary PWM and dead-time.
General timers	1 × 32-bit (TIMER1) + up to **9 × 16-bit** (TIMER2–4, 8–13)	2 × 32-bit (TIM2, TIM5) + multiple 16-bit general timers (TIM3,4,9–14)	near	Both have a rich mix of 16/32-bit timers for capture/compare, PWM, encoders.
Basic timers	**2 × basic** (TIMER5,6)	**2 × basic** (TIM6,7)	near	Similar simple timers used as time base or DAC triggers.
High-resolution timer	**SHRTIMER** dedicated high-res timer for power control	None in F4; HRTIM appears only on some STM32G4/F3	not near	GD32 has integrated high-res timer optimized for power/motor; F429 lacks equivalent HRTIM/SHRTIMER.
Watchdogs	Independent + Window	Independent + Window	near	Same watchdog schemes available.
RTC	32-bit calendar with alarm, LXTAL/LS clock	Similar RTC block with alarm, backup domain	near	Both have comparable RTC functionality.

5

Communication & Connectivity

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
USART/UART	**4 × USART** (0–2,5) + **2 × UART** (3–4) up to 22.5 Mbit/s	**4 × USART** (1–3,6) + **2 × UART** (4,5)	near	Very similar mix of USART/UART peripherals and capabilities.
I ² C	**3 × I ² C** (0–2), up to 1 MHz FM+	**3 × I ² C**	near	1:1 match in number and capability (Standard/Fast/Fast+).
SPI	**3 × SPI** (1–2), up to 192 kHz audio	Up to **6 × SPI/I ² S**	near	STM32F429 has more SPI/I ² S instances; GD32 offers Quad-SPI on SPI0.
I ² S	**2 × I ² S** (1–2), up to 192 kHz audio	Multiple I ² S (on SPI2,3,4,5)	near	Both can handle stereo audio streams; STM32 has more instances.
USB	**1 × USB OTG HS** with **embedded HS PHY** (HS/FS/OTG)	**1 × USB OTG FS** (internal PHY) + **1 × USB OTG HS** (requires external ULPI PHY)	near	Both offer OTG FS and HS connectivity; GD32 integrates HS PHY, STM32 uses external HS PHY on HS line.
Ethernet MAC	**10/100 MAC** with IEEE 1588 PTP	**10/100 MAC** with IEEE 1588 PTP	near	Very comparable Ethernet hardware, both with IEEE 1588 timestamping.
CAN	**3 × CAN 2.0A/B**	**2 × CAN 2.0A/B**	near	GD32 offers one extra CAN; otherwise CAN feature set is similar.
Other buses	EXMC for external memory, SQPI, CTC	FMC for external memory, Quad-SPI, SDIO, DCMI, etc.	near	Both target connectivity-rich applications; STM32 adds camera/SDIO, GD32 adds SQPI/TMU/SHRTIMER.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

6 External Memory & High-Speed Interfaces

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
Parallel memory IF	**EXMC** supporting SRAM/PSRAM, NOR, NAND, PC Card	**FMC** supporting SRAM/PSRAM, NOR, NAND, SDRAM	near	Both have flexible external memory controllers; STM32 adds SDRAM support.
Serial/Quad IF	**SQPI** (serial/dual/quad PSRAM/Flash mapped to 0xB0000000)	**Quad-SPJ** (for serial Flash, XIP)	near	Both can map external QSPI memories into address space.
Max ext. clock	EXMC clk up to HCLK/4 typical, detailed timing in datasheet	FMC up to ~90 MHz depending on timing configuration	near	External parallel interface speeds are in comparable range.

7 GPIO & Package

Criterion	GD32E507ZE (LQFP144)	STM32F429ZI (LQFP144, NUCLEO-F429ZI)	Tag	Reason
Package	**LQFP144**, 0.5 mm pitch	**LQFP144**, 0.5 mm pitch	near	Same package class; similar mechanical integration.
GPIO count	Up to **112 GPIOs**	Up to **114 GPIOs**	near	Both expose >100 GPIOs with rich AF multiplexing.
5 V tolerance	Many pins 5 V tolerant (see datasheet per pin)	Many pins 5 V tolerant (depending on I/O bank)	near	Voltage-tolerant input capability on both families.

8 GPIO & Package

Criterion	GD32E507ZE (LQFP144)	STM32F429ZI (LQFP144, NUCLEO-F429ZI)	Tag	Reason
Package	**LQFP144**, 0.5 mm pitch	**LQFP144**, 0.5 mm pitch	near	Same package class; similar mechanical integration.
GPIO count	Up to **112 GPIOs**	Up to **114 GPIOs**	near	Both expose >100 GPIOs with rich AF multiplexing.
5 V tolerance	Many pins 5 V tolerant (see datasheet per pin)	Many pins 5 V tolerant (depending on I/O bank)	near	Voltage-tolerant input capability on both families.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

9

Core & Memory

Criterion	GD32E507ZE – GD32E507Z-EVAL	STM32F429ZI – NUCLEO-F429ZI	Tag	Reason
RTOS support	Runs FreeRTOS or similar; integration mostly manual (projects/examples exist but less standardized).	Runs FreeRTOS; **CubeMX/IDE** generate RTOS projects with templates.	near / not near	Both can run RTOS; ST has much stronger automated tooling.
RTOS capacity	180 MHz, 512 KB Flash, 128 KB RAM – OK for RTOS + 1–2 middleware stacks.	180 MHz, 2 MB Flash, 256 KB RAM – comfortable for RTOS + multiple stacks.	near	Same performance class; STM32 has more memory headroom.
USB hardware	1× USB OTG HS with **embedded HS PHY** (HS/FS/OTG).	1× OTG FS (internal PHY) + 1× OTG HS (needs external PHY).	near	Both support FS/HS OTG; GD32 integrates HS PHY.
USB middleware	GD32 USB device/host library (CDC, MSC, HID, etc.), fewer canned RTOS combos.	CubeF4 USB Device/Host libraries with many ready examples and RTOS demos.	near / not near	Capabilities similar; ST ecosystem is richer and more turnkey.
File system	Uses **FatFS** on SPI/EXMC/SQPI Flash or SD via SPI/external SDIO; examples available but more manual.	**FatFS** integrated in CubeF4 with SDIO/SD, USB MSC, QSPI/NOR examples and RTOS variants.	near / not near	Both use FatFS; ST provides many official, combined templates.
TCP/IP (Ethernet)	ENET + lwIP possible; examples exist but fewer, typically hand-integrated with RTOS/FS.	ENET + **lwIP** in CubeF4 with HTTP server and FreeRTOS templates.	near / not near	Both use lwIP; ST gives more complete reference projects.



GD32E507ZE vs STM32F429ZI (NUCLEO-F429ZI)

10

Accelerators, Security & Special IP

Criterion	GD32E507ZE	STM32F429ZI	Tag	Reason
Trig/math accelerator	**TMU** (hardware trig/sqrt/atan, etc.)	None (use FPU + CMSIS-DSP in software)	not near	GD32 has a dedicated TMU; STM32F429 relies on software libraries.
High-res power timer	**SHRTIMER**	None in F4 series	not near	GD32 integrates a specialized power/motor timer comparable to STM32G4/F3 HRTIM, not present in F429.
Security/crypto	Basic (no explicit crypto mentioned)	Some variants include CRYPT and HASH accelerators	not near	STM32F429 typically has symmetric crypto accelerators; GD32E507 focuses more on control/motor side.
TrustZone	Cortex-M33 core supports TrustZone (if used)	Cortex-M4 has **no TrustZone**	not near	Different security model and privilege separation capabilities.



Board Equivalence Summary

For each non-ST evaluation board, a closest STM32 counterpart has been identified to enable fair cross-vendor comparison:

NXP MIMXRT1060-EVKC (i.MX RT1062, Cortex-M7 crossover) ↔ SM32H735G-DK (Cortex-M7, high-end, Ethernet + graphics)

NXP FRDM-MCXA156 (Cortex-M33, 96 MHz, low-power/industrial) ↔ NUCLEO-L552ZE-Q (Cortex-M33, low-power, similar Flash/RAM and peripherals)

Renesas FPB-RA0E1 (Cortex-M23, ultra-low-end) ↔ NUCLEO-L031K6 (Cortex-M0+, similar speed and small memory class)

Renesas FPB-RA8E1 (Cortex-M85, high-end control) ↔ NUCLEO-H743ZI2 (Cortex-M7, 400 MHz, large Flash/SRAM)

TI LP-MSPM0C1104 (Cortex-M0+, 16 KB / 1 KB, very low-end) ↔ NUCLEO-L031K6 (Cortex-M0+, 32 KB / 8 KB, same segment)

GigaDevice GD32E507Z-EVAL (GD32E507ZE, Cortex-M33, 180 MHz, high-performance control) ↔ NUCLEO-F429ZI (STM32F429ZI, Cortex-M4, 180 MHz, similar analog, timers, Ethernet, USB, external memory)



Our technology starts with You



Find out more at www.st.com

© STMicroelectronics - All rights reserved.

ST logo is a trademark or a registered trademark of STMicroelectronics International NV or its affiliates in the EU and/or other countries.

For additional information about ST trademarks, please refer to www.st.com/trademarks.

All other product or service names are the property of their respective owners.



For further support in creating a PowerPoint presentation, including graphic assets, formatting tools and additional information on the ST brand **you can visit the ST Brand Portal** <https://brandportal.st.com>

